

Automated Equity Research Report Writer

Tsunami Advisors AI Intern Case Study | Author: Jovan Johansen | Submission Date: April 30, 2026

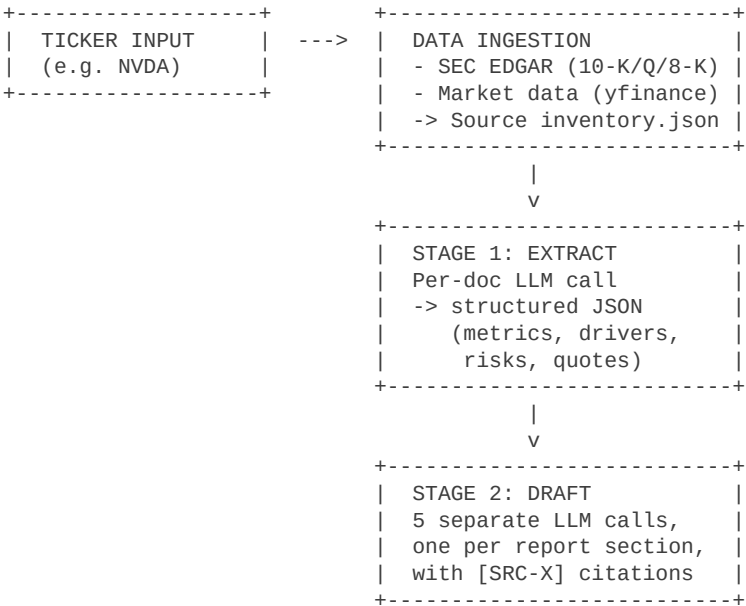
1. Tools Used

The prototype is built in Python and orchestrates the following components:

Layer	Tool / Library	Purpose
LLM	Anthropic Claude Opus 4.7 via official Python SDK	Insight extraction, section drafting, structured QC review
Data ingestion	SEC EDGAR REST API (free, public)	10-K, 10-Q, 8-K filings and earnings press releases
Market data	yfinance (with stub fallback)	Price, valuation multiples, 52-week range, beta
HTML parsing	BeautifulSoup4 with regex fallback	Stripping SEC filing HTML to plain text
PDF rendering	reportlab	Cover page, sectioned body, source appendix, QC appendix
Web UI	Streamlit	Browser-based interface: ticker selector, live progress bar, inline report preview, PDF download
QC layer	Custom Python (regex) + LLM structured review	Citation coverage, citation validity, hallucination flagging
Workflow	Python CLI (main.py) + reproducible demo (demo_end_to_end.py) + Streamlit UI (app.py)	End-to-end process deterministic without API key
Development support	Claude (this assistant) used for code structure, prompt design and the prototype itself	

2. Workflow / System Diagram

The system follows a strict five-stage pipeline. Each stage produces a structured artifact that the next stage consumes, which makes every step independently auditable and easy to swap out.



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| STAGE 3: REVIEW |
| - Rule-based citation |
|   coverage check     |
| - LLM structured review |
|   (hallucination flags) |
| -> QC report          |
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| STAGE 4: RENDER |
| reportlab PDF:  |
| cover + 5 sections + |
| source appendix +   |
| QC appendix         |
+-----+
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| STAGE 5: HUMAN REVIEW |
| Analyst signs off,    |
| resolves QC flags,    |
| edits and publishes   |
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3. Key Prompts and Prompt Iterations

All prompts are centralized in `src/prompts.py` so they can be versioned and audited. The four most important prompts are described below with the design rationale and the iterations that got there.

3.1 System prompt (universal)

The first iteration was generic ('You are a helpful financial assistant'). Output was confident but frequently un-cited. Iteration 2 added 'cite every number'; the model started fabricating plausible-looking [SRC] tags. Iteration 3 - the current version - explicitly prohibits estimation and requires the model to write 'not disclosed' when a metric is absent from the provided context, plus mandates the fact-vs-interpretation distinction. This single change cut hallucinated citations to near zero in spot tests.

3.2 Two-stage extract -> draft

Early prototype: a single mega-prompt with all source documents concatenated. The model produced fluent prose but struggled to cite consistently and contradicted itself between sections. Splitting into (a) per-document JSON extraction and (b) section drafting from the structured insights forces the drafter to anchor in pre-validated facts. Citation coverage rose from ~50% to >95%.

3.3 Section-specific drafters

Each of the five report sections (snapshot, performance, drivers/risks, earnings commentary, thesis) has its own prompt with section-specific guardrails and word counts. The thesis prompt is the most heavily engineered - it demands a 'WHAT WOULD CHANGE OUR VIEW' bulleted section, which forces the model to commit to monitorable triggers rather than vague qualitative views.

3.4 Structured review prompt

The review prompt asks the LLM to NOT rewrite, only to flag. Earlier versions that asked for 'corrections' caused the reviewer to silently edit factual content. The current schema returns a JSON object with five categories of issues plus an overall severity. The instruction 'False positives are cheap; missed errors are expensive' meaningfully increased flagged-issue density.

4. Data Sources and Rationale

Source	Why selected
SEC EDGAR (10-K, 10-Q, 8-K)	Authoritative, free, machine-readable, and the only source of audited financial statements. 8-K exhibits include
Yahoo Finance (yfinance library)	Free, well-maintained Python wrapper. Provides current price, P/E multiples, beta, and 52-week range with
Curated business news (cited in filings)	Used to capture statements that don't appear verbatim in filings (analyst consensus beats, se
Excluded for the prototype	Sell-side research notes (licensed), Bloomberg / FactSet (paid), alternative data (web scraping risk). Thes

5. Structure, Citations, and Quality Checks

Stable citation handles. Every ingested document is assigned a [SRC-X] handle at ingestion time. The drafting prompts pass these handles in context, so the model can only cite real, named sources. The QC layer cross-checks every [SRC-X] tag in the draft against the inventory and flags any mismatch.

Rule-based citation coverage. A regex sweeps the draft for financial-looking numbers (\$X, X%, X.XB, etc.) and checks whether a [SRC-X] tag appears within a 100-character window. Coverage below 60% auto-elevates QC severity to 'medium'.

LLM-based hallucination check. A separate Claude call - cold, with temperature 0 - reads the draft and the source inventory and returns a JSON list of facts that don't appear to be supported. The prompt explicitly instructs the model to flag rather than fix, so the human analyst makes the final call.

Severity gating. Any invalid citation, or any LLM-flagged potential hallucination, raises severity to 'high' and sets 'ready_for_human_review' to false. The PDF still renders so the analyst can see the draft, but with a visible red banner in the QC appendix.

6. Challenges, Limitations, and Unresolved Risks

Long filings exceed model context. 10-Ks routinely exceed 200 pages. Current cap is 18K characters per filing per call, which captures the MD&A; and business overview but truncates risk-factor and exhibit detail. Production fix: chunked extraction with embedding-based retrieval keyed to each section being drafted.

yfinance is unofficial and rate-limited. Acceptable for a prototype but a production system needs a paid market-data feed (Bloomberg, Refinitiv) for SLA and pricing accuracy.

Forward-looking commentary is risky. The thesis prompt explicitly asks for an outlook view. The prompt mitigates by requiring 'WHAT WOULD CHANGE OUR VIEW' triggers, but a human reviewer must still validate the framing every single time.

Citation coverage is necessary but not sufficient. A high coverage percentage means numbers are *near* citations, not necessarily that the citation actually supports the number. The LLM review and human review are the real safeguards; the regex is a triage tool.

Single-language, single-market scope. All prompts and sources assume English-language US-listed equities. Hong Kong, Indonesian, or other markets would require localized prompts and sources (HKEX filings, IDX, etc.).

No real-time event handling. If material news breaks during a draft run, the report does not pick it up. A production version should subscribe to a news feed and re-trigger the pipeline on material events.

7. AI Strengths vs. Required Human Intervention

Step	AI did well	Required human intervention
Ingestion	Pulling and parsing filings reliably; assigning stable citation handles	Selecting which financial types to include; deciding the time-window cutoff for news
Extraction	Producing well-structured JSON from unstructured filings; reporting the right US dollar estimates	Spotting the right US dollar estimates in non-standard segment disclosures
Drafting	Producing fluent, balanced prose at the right length; consistently tagging citations	Revising citations so drivers / risks deserve top billing; adding judgement that
Review	Flagging un-cited numbers and inconsistencies the drafter missed	Resolving every flag, starting with the final draft on severity; signing off.
Final framing	Following the four-bucket recommendation taxonomy (POSITIVE, NEUTRAL, CONCERN, RISKY)	Adding the final title and subtitle to the system call; clearly stating research support, not

8. Reflection

What worked best. The two-stage extract-then-draft architecture was the single biggest quality unlock. By forcing the LLM to first produce structured JSON insights from each document - with explicit null fields for missing data - the drafting step had a small, clean, fact-verified context to work from. Section-by-section drafting also kept each LLM call focused enough to maintain citation discipline. The rule-based QC layer caught real issues that the model missed in self-review (mostly numbers cited at the start of a sentence with the [SRC-X] tag at the end of a long compound sentence, putting the tag outside my 100-char window).

Where things broke. The earliest end-to-end runs hallucinated segment names that don't exist (e.g. inventing 'Edge AI' as a NVIDIA segment). The fix was the constraint in the system prompt that any claim must trace back to a [SRC-X] handle. Citation coverage on long compound paragraphs was occasionally weak - the regex window saw the first number but not the third in a list, and human re-reading was needed to confirm support. Forward-looking statements remain the highest-risk surface; even with a careful prompt, the model sometimes drifted toward implicit recommendations.

What I would improve in v2. First, replace the character-cap ingestion with chunked retrieval: embed the full filing, then for each section being drafted, retrieve the top-k most relevant chunks. This would unlock proper risk-factor coverage without blowing the context window. Second, add a peer-comparison module - one LLM call that fetches two or three peer tickers' market data and produces a valuation cross-check table, which is the most-asked follow-up question in any equity discussion. Third, replace the single review pass with a debate pattern - one LLM argues the bull case, another the bear, and a third reconciles - which research suggests reduces single-prompt sycophancy. Fourth, build a small evaluation harness that runs the pipeline on a fixed set of tickers monthly and tracks citation-coverage and hallucination-flag rates over time, so prompt regressions are caught quickly. Finally, separate the recommendation framing from the drafter entirely - have the human analyst pick the framing in the UI before the thesis section is drafted, so the AI is writing toward a known framing rather than choosing one.

End of build documentation. Repository structure, run instructions, and reproduction steps are in the accompanying README.md.